

# ISHRAT BHULLAR

AI Engineer • Retrieval-Augmented Generation & Backend Systems

ibhullar\_be22@thapar.edu | linkedin.com/in/ishrat-bhullar-4a1441295 | github.com/Ishrat-Bhullar | Mohali, India

## SUMMARY

Computer Engineering student and six-month AI Engineering Intern at EY, focused on retrieval-augmented generation, multi-agent AI systems, and FastAPI backends for enterprise AI. Built document-intelligence systems that return source-traceable answers over private corpora — including scanned and non-English material — and a nine-agent platform that produces software engineering artifacts under human approval. Comfortable owning a feature end to end, from retrieval design through REST API to React interface, with LLMs deployed entirely on-premise for clients who cannot use external APIs.

## EXPERIENCE

### AI Engineering Intern — EY (Ernst & Young)

Jan 2026 – Jul 2026

Technology Consulting, CNS — Technology Strategy & Transformation. GenAI systems for enterprise and government clients.

Python • FastAPI • React • PostgreSQL • ChromaDB • FAISS • BM25 • Ollama • Docker

- Built a **keyword-only retrieval baseline** (BM25 with cross-encoder reranking) before adopting a vector database, to test whether an air-gapped deployment could avoid embedding infrastructure; evaluation isolated the exact failure modes — paraphrased queries, indirect references, non-English content — that justified the redesign.
- Re-architected retrieval as a dual path, merging FAISS dense search with BM25 lexical scoring through **Reciprocal Rank Fusion**, recovering exact-term matches dense-only search missed while retaining semantic recall on paraphrased queries.
- Extended coverage with **multilingual E5 embeddings**, letting a single index serve cross-lingual search without separate per-language stores, and an **OCR ingestion pipeline** that converts scanned documents into chunked, metadata-tagged embeddings, replacing a manual update process.
- Served retrieval, inference, and ingestion through FastAPI REST endpoints connecting a React chat interface to a PostgreSQL store, with open-source LLMs deployed on-premise through **Ollama** to satisfy client data-residency rules and remove external API dependency.
- Cut end-to-end answer latency from **~60 seconds to 10–15 seconds** by replacing dense-only vector retrieval with the hybrid path, switching the generation model, adding query rewriting, and tuning decoding parameters.
- Grounded responses in retrieved text through inline citation generation, mapping each claim to its source document and page so answers stay auditable.

## PROJECTS

### Multi-Agent Autonomous SDLC Platform — EY Internship Project

2026

Python • FastAPI • React • TypeScript • PostgreSQL • LangGraph • LLM Agents

- Generalised the grounding-and-human-oversight pattern proven across both retrieval systems into a **nine-agent pipeline** that converts a plain-language business brief into requirements, user stories, solution architecture, database schemas, UI/UX designs, security and compliance assessments, and technical documentation.
- Sequenced agent handoffs through a stateful orchestrator with two mandatory **human approval gates**, so a reviewer can approve, reject, or regenerate any agent's output before the next stage runs.
- Centralised project context in a memory agent that every downstream agent reads from and writes to, preventing the cross-artifact inconsistency that arises when each agent re-derives requirements independently.
- Surfaced agent status, generated artifacts, and approval history in a React dashboard, persisting full run state in PostgreSQL so executions stay auditable and replayable.

### Maharashtra Government Environmental Clearance AI Portal — EY Client Project — Testing & Validation

2026

OCR • Document Classification • AI Output Verification • Regression Testing

- Validated a live government document-evaluation platform ahead of deployment: functional testing across a twelve-stage pipeline, end-to-end workflow validation, and verification of AI-extracted parameters and generated reports against the source documents.
- Assessed OCR extraction quality across scanned document types, logged and tracked defects with the development team, and ran regression testing to confirm fixes held — establishing the **source-traceability and human-checkpoint standards** carried into every system I built afterwards.

### Autonomous Water-Cleaning Surface Vehicle — Capstone Project

2025

Arduino • Embedded C/C++ • IR & Ultrasonic Sensors • IoT • Blockchain

- Engineered an Arduino-controlled surface vehicle that navigates autonomously and triggers filtration from combined IR and ultrasonic input, logging sensor telemetry to a tamper-evident blockchain ledger and exposing live status, cycle history, and threshold alerts through a web dashboard.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, C++

**Frameworks:** FastAPI, React, Streamlit, LangGraph

**AI / LLMs:** Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agent Orchestration, Prompt Engineering, LLM Agents, Ollama (local LLM deployment)

**Retrieval:** Hybrid Search, BM25, Reciprocal Rank Fusion, Cross-Encoder Reranking, Semantic Search, Multilingual Embeddings (E5), Chunking Strategies, Citation Grounding, OCR Ingestion, Document Intelligence, Information Retrieval

**Backend & Data:** REST API Design, Async Services, PostgreSQL, ChromaDB, FAISS, On-Premise Inference

**Dev Tools:** Git, GitHub, Docker, Linux

## EDUCATION

### B.E. in Computer Engineering — Thapar Institute of Engineering & Technology, Patiala

2022 – 2026

CGPA 7.84 / 10 • Sacred Heart Sr. Sec. School, Chandigarh — CBSE Class XII (Non-Medical): 93.4%

## LEADERSHIP & INVOLVEMENT

Core team, five student organisations • Graphic designer, **Echoes** magazine • HSBC Case Program • UNICEF U-Report • NSS 120+ hrs